Experiment Design Plan

Multi-Agent Reinforcement Learning for Adaptive Traffic Signal Optimization in Smart Cities

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1. Objective

The objective of this experiment design is to **systematically evaluate** the effectiveness of a Multi-Agent Reinforcement Learning (MARL) based traffic signal control system against a traditional fixed-time baseline.

The experiments aim to:

- Quantify performance improvements
- Analyze agent behavior under varying traffic conditions
- Evaluate robustness and generalization
- Validate feasibility of decentralized control

2. Experimental Setup

2.1 Environment Configuration

- Grid-based traffic network
- Initial size: **4 × 4 intersections**
- Four მიმართულ traffic flow (North, South, East, West)
- Discrete-time simulation steps

2.2 Systems Under Comparison

1. Baseline System

- Fixed-time signal control
- Predefined signal cycles
- No real-time adaptation

2. MARL-Based System (In Progress)

- Decentralized agents per intersection
- Adaptive signal control using reinforcement learning

3. Independent Variables

The experiments will vary the following parameters:

3.1 Traffic Density

- Low traffic
- Medium traffic
- High traffic

3.2 Traffic Patterns

- Uniform traffic distribution
- Directional bias (e.g., peak flow in one direction)
- Dynamic traffic changes over time

3.3 Network Complexity (Later Phase)

- 4×4 grid (initial)
- 6×6 / 8×8 grid (scalability phase)

4. Dependent Variables (Metrics)

The following performance metrics will be measured:

4.1 Average Waiting Time

- Mean waiting time per vehicle

4.2 Queue Length

- Average queue length per intersection

4.3 Throughput

- Number of vehicles processed per unit time

4.4 Congestion Index

- Ratio of stopped vehicles to total vehicles

5. Experimental Scenarios

Scenario 1 - Baseline Validation

- Run fixed-time system across all traffic densities
- Validate simulation consistency

Scenario 2 - MARL Performance (Initial)

- Train MARL agents on 4×4 grid
- Compare against baseline

Scenario 3 - Generalization Test

- Train under one traffic condition
- Test under different conditions

Scenario 4 - Dynamic Traffic Behavior

- Introduce fluctuating traffic patterns
- Evaluate adaptability of MARL agents

Scenario 5 - Controlled Scalability (Later Phase)

- Increase grid size
- Measure performance stability

6. Training Configuration (MARL)

6.1 Episodes

- Each episode represents a traffic simulation run

6.2 Training Duration

- Fixed number of episodes (to be tuned)

6.3 Exploration Strategy

- ε-greedy or equivalent strategy

6.4 Convergence Criteria

- Stabilization of reward
- Consistent performance across episodes

7. Data Collection

During each experiment, the following data will be recorded:

- Vehicle-level metrics (waiting time, movement)
- Intersection-level metrics (queue length, signal states)
- System-level metrics (throughput, congestion)

All results will be stored in structured logs for analysis.

8. Analysis Methodology

8.1 Comparative Analysis

- Compare MARL vs baseline across all metrics

8.2 Statistical Analysis

- Average performance over multiple runs
- Variance and stability analysis

8.3 Visualization

- Time-series graphs
- Heatmaps of congestion
- Performance comparison charts

9. Expected Outcomes

The MARL system is expected to:

- Reduce average waiting time
- Reduce queue lengths
- Improve overall throughput
- Adapt better to dynamic traffic conditions

10. Validity Considerations

Internal Validity

- Controlled simulation environment ensures consistency

External Validity

- Real-world applicability may be limited in early stages

Limitations

- Simplified traffic assumptions
- Lack of real-world data integration (current phase)

11. Reproducibility

To ensure reproducibility:

- Simulation parameters will be documented
- Random seeds will be controlled
- Experiment configurations will be versioned

12. Summary

This experiment design provides a **structured and reproducible framework** to evaluate the effectiveness of MARL in traffic signal optimization.

By systematically varying traffic conditions and comparing against a baseline system, the study aims to generate **credible, measurable, and research-grade results**.